

Tezpur University

Mid. Semester Exams. — Spring 2025

CS417 : OPERATING SYSTEM

Max. Marks: 30

Max. Time: $1\frac{1}{2}$ hrs.*All questions are compulsory except otherwise stated.**Figures in the right margin indicate mark(s) allotted to the individual/groups of question(s).*

1. Given the requirements of a system below, answer the following questions.

Process	Arrival Time	Burst Time
P_0	0	5
P_1	1	3
P_2	2	3
P_3	4	1
P_4	5	2

- (a) Find the average *turn-around-time* (TAT) in preemptive shortest remaining time next CPU scheduling algorithm, . 1½
- (b) Find the average TAT in non-preemptive shortest job first, and 1½
- (c) Give a comparative statement on the average TAT with proper justification. 2
2. We have a system as depicted below. Can a request of (0, 4, 2, 0) for the resources A, B, C and D, respectively, from P_1 be immediately satisfied? Substantiate your answer(s). 3 + 2 = 5

	Allocation	Max	Available
	ABCD	ABCD	ABCD
P_0	0 0 1 2	0 0 1 2	1 1 0 0
P_1	1 4 2 0	1 7 5 0	
P_2	1 3 5 4	2 3 5 6	
P_3	0 6 3 2	0 6 5 2	
P_4	0 0 1 4	0 6 5 6	

3. Write the Bakery algorithm, multiple process solution to the *critical section* problem. 3
4. Consider a priority CPU scheduling algorithm which may be called as *longest remaining time first* (LTRF) that schedules CPU based on the longest remaining time. The system model is a three process system P_0 through P_2 where the subscripted number reflects the process ID. If 2, 4 and 8 are the CPU Burst time (in ms) required by the processes P_0 through P_2 , respectively, what will be the overhead of the system if context-switching takes 100ns. Consider all the processes arrived at the same time and any tie is broken by giving priority to the process with lower process ID. 2
5. What are batched systems? Identify a problem, cost vis-à-vis usage, that a batch system pose. Suggest a solution. 1 + 2 + 1 = 4

6. Comment on process creation in terms of: (a) resource sharing, (b) execution, and (c) address space. If possible substantiate your answer with ample example(s). 3
7. Given the producer-consumer problem of cooperative processes and the implementation of it using IPC shared memory. The producer and the consumer are guarded from accessing the shared memory segment through a shared flag as in the following C program snippet. What can happen in this scenario and how can you prevent it? Provide a code or an algorithm for your solution. $2 + 2 + 1 = 5$

```
/* share stuffs */
#define SHM_SZ 1024 // size of the shared memory segment

struct shared_stuff {
    int dirty_flag;
    char some_text[SHM_SZ]

/* server */
int shmid = shmget(key, SHM_SZ, IPC_CREAT | 0666); // create a shared mem.
struct shared_stuff *memo = shmat(shmid, NULL, 0); // now attach to it
memo->dirty_flag = 1; // signal client: server is waiting
while (running) {
    while (memo->dirty_flag == 1) { // wait till the client clears the
        sleep(1); // dirty flag to signal its ready
    }
    /**
    * do some stuff: write access the shared memory
    * or prepare to break out of the communication
    */
    memo->dirty_flag = 1; // indicate something is in the shared memory
    // if necessary break out of the IPC here
}
```

8. Prove that Peterson's solution to critical-section problem addresses all the three criteria: mutual exclusion, progress, and bounded waiting. 3

OR

Show a solution to dining philosopher's problem with the swap hardware solution. 3